DrivePulse Car Rental Management

Complete System Architecture, Operational Workflow & $0 Production Guide

Target: Internal Administrative & Operations Staff | Statutory Indian GST | Responsive Mobile & Desktop Web App

1. Executive Overview & Core Business Model

Engineered strictly for internal fleet managers and counter staff.

- Admin-Only Access: Strictly internal operations hub (Super Admin, Operations Manager, Counter Staff).
- Frictionless Customer Walk-ins: Staff can type customer details on the fly or pick from registered directory.
- People-Based Dynamic Rent: No rigid fixed rates; rent is dynamically negotiated based on passenger count and route.
- Clean Fleet View: Odometer readings are captured strictly during dispatch/return inspections, eliminating clutter.
- Road Legality & Safety Gate: Vehicles cannot be dispatched if Insurance, Fitness, or PUC are expired (Manager override required).
- Statutory Indian GST Compliance: Full 18% GST support (9% CGST + 9% SGST), SAC Code 996601, and official PDF Tax Invoices.

2. Full-Stack Technology Stack & Architecture

End-to-End Architectural Components

- Frontend SPA: React 18 + TypeScript + Vite + Tailwind CSS + Lucide Icons + TanStack Table + Signature Canvas
- Backend API: Node.js + Express + TypeScript + Zod Validation + Helmet + dual JWT Cookies (Access + Refresh)
- Database & ORM: Prisma ORM with 18 relational models (SQLite for local dev, PostgreSQL for production)
- PDF Engine: PDFKit vector document generator for Rental Agreements, GST Invoices, and Payment Receipts
- Mobile Adaptation: Off-canvas sliding drawer, sticky bottom navigation bar, touch canvas, and responsive mobile cards

SYSTEM ARCHITECTURE MATRIX

Layer	Technology	Local Development	Production ($0 Cloud Tier)
Client (SPA)	React 18 + Vite + TS	http://localhost:3000	Vercel / Render Static CDN
API Server	Express + TypeScript + Zod	http://localhost:5000	Render Free Web Service
Database	Prisma ORM	SQLite (dev.db)	Neon Serverless PostgreSQL
Storage / PDFs	Local FS + PDFKit	server/uploads	Local Persistent / Cloudinary

3. Mobile & Tablet Optimization Architecture

Responsive design for staff using smartphones, iPads, and counter tablets.

> **Why Mobile Optimization Matters for Rental Operations**
> Rental staff inspect cars in parking lots, take photos on mobile devices, collect customer signatures at the car door, and record fast check-ins on mobile phones. DrivePulse is fully responsive with native touch optimizations on smartphones (iOS / Android) and tablets.

Key Mobile Features Implemented:

1. Mobile Slide-Over Drawer: Off-canvas navigation menu with backdrop blur, dark/light theme switcher, and staff profile card.
2. Sticky Mobile Bottom Navigation: Instant one-thumb access to Dashboard, Fleet, Bookings, and an elevated "New Booking" button.
3. Touch Signature Canvas: Supports HTML5 touch events (onTouchStart/Move/End) with touch-action:none preventing page scrolling.
4. Mobile Live Quotation Bar: In /bookings/new, a sticky bottom bar displays real-time Total and Balance Due with "Review" and "Confirm" triggers.
5. Responsive Card Layout: On small screens (<768px), data tables automatically transform into touch-friendly cards with direct action buttons.
6. Anti-Zoom Form Controls: Form inputs are styled to prevent unwanted viewport zooming on iOS mobile devices.

4. Operational Workflows & Business Logic

Workflow 1: Frictionless Booking Creation

Staff navigate to /bookings/new. They choose between "Enter Manually" (typing Name, Phone, DL, City) or selecting an existing customer. The staff enters agreed rental rate based on passenger count (e.g., 5 passengers) without being constrained by fixed tariffs. Add-ons (child seats, chauffeur allowance, GPS) and courtesy discounts are computed live with 18% GST.

Workflow 2: Vehicle Handover & Digital Inspection

Before release, counter staff perform a digital handover inspection recording: starting odometer, fuel gauge level (0/4 to 4/4), inventory accessories (spare wheel, jack, toolkit, Fastag, RC card copy), pre-existing damages, and customer signature on canvas. Upon completion, the booking status automatically transitions to PICKED_UP.

Workflow 3: Return Check-in, Settlement & GST Invoicing

Upon vehicle return, staff record return odometer and fuel level. Late fees, extra km charges, or damage deductions are debited against the refundable security deposit. Cashiers record payments (UPI, Cash, Card, Bank Transfer) generating instant PDF receipts and official GST Tax Invoices with SAC 996601.

5. Zero-Cost ($0) Production Deployment Guide for Client Testing

Step-by-step instructions to run the live application for $0 with no credit card required.

100% Free Production Stack for Client Acceptance Testing (UAT)

Your client can test the full-stack system live in production on their phone and computer without incurring any cloud hosting costs. The entire stack leverages legitimate, enterprise-grade free tiers from Neon, Render, and Vercel.

ZERO-COST ($0) CLOUD HOSTING BREAKDOWN

Service Layer	Platform / Provider	Free Tier Limits	Testing Cost
Managed Database	Neon Serverless Postgres	0.5 GB SSD, auto-suspend, SSL	$0.00 / month
Backend API Server	Render Web Service	512 MB RAM, 750 free hrs/mo	$0.00 / month
Frontend Client (SPA)	Vercel / Render Static	Global Edge CDN, auto SSL	$0.00 / month
Total Cost for Client	Production Environment	Full mobile & desktop testing	**$0.00 (Zero Cost)**

Step-by-Step Live Deployment Instructions:

• Step 1: Create Neon PostgreSQL Database: Sign up free at neon.tech (no credit card needed). Create project "drivepulse-prod". Copy the connection string (DATABASE_URL).

• Step 2: Push Repository to GitHub: Push your clean git commits to a private or public GitHub repo.

• Step 3: Deploy Backend on Render: Sign up at render.com. Create "New Web Service" connecting your repo. Set root directory to "server", build command to "npm install && npm run build && npx prisma db push && npm run seed", start command to "npm start". Add env vars: DATABASE_URL, JWT_SECRET, JWT_REFRESH_SECRET, NODE_ENV=production, PORT=5000, CLIENT_URL=https://your-app.vercel.app.

• Step 4: Deploy Frontend on Vercel: Sign up at vercel.com. Import repo, set root directory to "client", framework preset to "Vite". Add env var VITE_API_BASE_URL pointing to your Render API. Deploy!

• Step 5: Send Live Testing Link to Client: Provide the client with the live HTTPS URL and pre-configured staff test credentials.

Client User Acceptance Testing (UAT) Verification Script

1. Open on Smartphone: Test the responsive bottom navigation bar, menu drawer, and card layouts.
2. Create Booking: Enter a customer manually, select vehicle, set negotiated rent based on passengers, review quote.
3. Dispatch Vehicle: Complete Handover inspection checklist and sign on touch canvas; status flips to PICKED_UP.
4. Verify Invoicing: Record advance payment, download official PDF receipt and statutory 18% GST Tax Invoice.

6. Role-Based Access Control (RBAC) & Security

PERMISSIONS MATRIX ACROSS STAFF ROLES

System Feature / Operation	Super Admin	Operations Manager	Counter Staff
View Fleet, Dashboard & Analytics	Full Access	Full Access	Operational View
Create Bookings & Record Walk-ins	Full Access	Full Access	Full Access
Handover & Return Inspections	Full Access	Full Access	Full Access
Collect Cashier Payments & Invoices	Full Access	Full Access	Full Access
Override Expired Compliance Docs	Authorized	Authorized	**BLOCKED (403)**
Bypass Outstanding Balance at Handover	Authorized	Authorized	**BLOCKED (403)**
Staff User Management & Audit Logs	Full Access	Audit Logs Only	**BLOCKED (403)**

7. Pre-Configured Test Accounts & Verification

- Super Admin Account: admin@drivepulse.com | Password: Password@123 (Full system privileges)
- Operations Manager Account: manager@drivepulse.com | Password: Password@123 (Compliance & financial overrides)
- Counter Staff Account: staff@drivepulse.com | Password: Password@123 (Front-desk bookings, inspections, cashier)

Quality Assurance Verification & Test Suite Results

- 29/29 Automated Integration Tests Passing (100% test pass rate in server/test/run-tests.ts)
- Zero Build / Type Errors: TypeScript strict mode enabled across both frontend and backend
- Mobile Browser Testing: Verified in mobile viewport (390x844) with bottom navigation and drawer
- Clean Workspace: Temporary files removed, production .gitignore configured, committed to git

DrivePulse Car Rental Management System — Developed for production excellence, compliance, mobile ease, and zero-cost client evaluation.